User Behavior & Conversion Analysis for Micro-Mobility Services

Prepared for: Director of Marketing and Cyclistic Executive Team

Project: Google Data Analytics Capstone Case Study — User Conversion Strategy

Tools Used: Google BigQuery (SQL), Looker Studio (Visualizations)

1. Executive Summary

This study analyzes 12 months of historical trip data from Cyclistic to identify behavioral differences between casual riders and annual members, aiming to design targeted marketing strategies for annual membership conversion.

Key Findings:

- **Usage Intent:** Members use Cyclistic primarily for daily commuting (Monday–Friday peaks during rush hours with short ~12-minute rides). Casual riders use the service for recreational/tourist purposes (weekend spikes with average rides around 39 minutes).
- **Seasonality:** Casual usage is heavily concentrated during summer months (May–September) and declines sharply in winter, whereas members maintain consistent baseline usage year-round.
- **Geographic Focus:** Casual trip origins are heavily concentrated in tourist and waterfront hubs (e.g., Streeter Dr & Grand Ave, Lake Shore Dr).

Strategic Recommendations:

1. **Summer Membership Credit:** Allow casual riders to apply summer single-pass purchases as a credit toward an annual membership.

2. **Geo-Targeted Marketing:** Focus physical and digital promotional efforts on the top 10 casual departure stations.

3. **Gamified Ride-Time Rewards:** Reward longer ride durations with points redeemable for membership discounts.

2. Phase 1: Ask

Context & Business Problem

Cyclistic operates a bike-share system in Chicago. The marketing strategy relies on maximizing annual memberships, as members generate significantly higher long-term profitability than casual riders. To drive growth, Cyclistic needs to convert casual riders into annual members.

Key Business Questions

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy a Cyclistic annual membership?
3. How can Cyclistic use digital media to influence casual riders to become members?

3. Phase 2: Prepare

Data Source & Rationale

Dataset: Divvy trip data provided by Motivate International Inc. under an open-source license.

Selection: A continuous 12-month historical period (Q2 2019 through Q1 2020) was selected to capture full seasonal trends and contemporary usage behaviors.

Data Storage & Ingestion

Files were uploaded to Google Cloud Storage (`gs://bucket-cyclistic/`) and ingested into BigQuery using `LOAD DATA OVERWRITE`.

```sql
SQL

-- Ingesting quarterly datasets into raw staging tables

LOAD DATA OVERWRITE
`project-941e89dd-df86-4a8c-ba4.Cyclistic.q1_2019`
FROM FILES (
  format = 'CSV',
  uris =
['gs://bucket-cyclistic/cyclistic_folder/divvy_trips_2019_q1.csv'],
  skip_leading_rows = 1,
  allow_quoted_newlines = TRUE
);

-- Note: Repeat LOAD DATA command for Q2 2019, Q3 2019, and Q4 2019
tables
```

4. Phase 3: Process

Schema Unification & Raw Data Consolidation

Due to schema changes across historical quarters (legacy column names vs. modern naming structure), fields were standardized and mapped into a single raw table: cyclistic_12_months_raw.

```sql
CREATE OR REPLACE TABLE
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_raw` AS

-- 1. Quarter 2 2019 (Q2)
SELECT
  CAST(`01 - Rental Details Rental Id` AS STRING) AS ride_id,
  CAST(`01 - Rental Details Local Start Time` AS TIMESTAMP) AS
started_at,
  CAST(`01 - Rental Details Local End Time` AS TIMESTAMP) AS
ended_at,
  `03 - Rental Start Station Name` AS start_station_name,
  `02 - Rental End Station Name` AS end_station_name,
  `User Type` AS member_casual
FROM `project-941e89dd-df86-4a8c-ba4.Cyclistic.q2_2019`

UNION ALL

-- 2. Quarter 3 2019 (Q3)
SELECT
  CAST(trip_id AS STRING) AS ride_id,
  CAST(start_time AS TIMESTAMP) AS started_at,
  CAST(end_time AS TIMESTAMP) AS ended_at,
  from_station_name AS start_station_name,
  to_station_name AS end_station_name,
  usertype AS member_casual
FROM `project-941e89dd-df86-4a8c-ba4.Cyclistic.q3_2019`

UNION ALL

-- 3. Quarter 4 2019 (Q4)
SELECT
  CAST(trip_id AS STRING) AS ride_id,
```

```
  CAST(start_time AS TIMESTAMP) AS started_at,
  CAST(end_time AS TIMESTAMP) AS ended_at,
  from_station_name AS start_station_name,
  to_station_name AS end_station_name,
  usertype AS member_casual
FROM `project-941e89dd-df86-4a8c-ba4.Cyclistic.q4_2019`

UNION ALL

-- 4. Quarter 1 2020 (Q1 2020)
SELECT
  CAST(ride_id AS STRING) AS ride_id,
  CAST(started_at AS TIMESTAMP) AS started_at,
  CAST(ended_at AS TIMESTAMP) AS ended_at,
  start_station_name,
  end_station_name,
  member_casual
FROM `project-941e89dd-df86-4a8c-ba4.Cyclistic.q1_2020`;
```

Data Cleaning & Transformation

The raw consolidated table was cleaned to build `cyclistic_12_months_clean`:

1. **User Naming Standardization:** Standardized legacy labels (`Subscriber` → `member`, `Customer` → `casual`).
2. **Metrics & Date Dimension Derivation:** Calculated `ride_length_minutes`, `day_of_week`, `day_name`, `month`, and `year`.
3. **Quality Filtering:** Removed records with missing station names/timestamps, false starts (< 1 minute), clock anomalies ($\le$ 0 minutes), and system maintenance outliers ($\ge$ 24 hours).

```SQL
CREATE OR REPLACE TABLE
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
AS
SELECT
  ride_id,
  -- 1. Standardize user type
  CASE
```

```
    WHEN LOWER(member_casual) IN ('subscriber', 'member') THEN
'member'
    WHEN LOWER(member_casual) IN ('customer', 'casual') THEN 'casual'
    ELSE member_casual
  END AS member_casual,
  started_at,
  ended_at,
  -- 2. Derive ride length duration
  TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS
ride_length_minutes,
  TIMESTAMP_DIFF(ended_at, started_at, SECOND) AS
ride_length_seconds,
  -- 3. Extract temporal fields
  EXTRACT(DAYOFWEEK FROM started_at) AS day_of_week, -- 1=Sunday,
7=Saturday
  FORMAT_TIMESTAMP('%A', started_at) AS day_name,
  EXTRACT(MONTH FROM started_at) AS month,
  FORMAT_TIMESTAMP('%B', started_at) AS month_name,
  EXTRACT(YEAR FROM started_at) AS year,
  -- 4. Station details
  start_station_name,
  end_station_name
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_raw`
-- 5. Data quality filters
WHERE
  started_at IS NOT NULL
  AND ended_at IS NOT NULL
  AND ended_at > started_at
  AND TIMESTAMP_DIFF(ended_at, started_at, MINUTE) >= 1   -- Exclude
< 1 min (false starts)
  AND TIMESTAMP_DIFF(ended_at, started_at, HOUR) < 24    -- Exclude
>= 24 hrs (maintenance/loss)
  AND start_station_name IS NOT NULL
  AND end_station_name IS NOT NULL;
```

Cleaning Verification

A quality check was executed to confirm correct data aggregation and verify expected baseline volume metrics:

```sql
SQL
SELECT
  member_casual,
  COUNT(*) AS total_trips,
  ROUND(AVG(ride_length_minutes), 2) AS avg_duration_min
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
GROUP BY member_casual;
```

5. Phase 4: Analyze

Key Comparative Summary

Metric / Dimension	**Annual Members (member)**	**Casual Riders (casual)**
Primary Usage Purpose	Commuting / Utility Transport	Leisure, Tourism & Recreation
Peak Days of Activity	Monday – Friday (Workdays)	Saturday & Sunday (Weekends)
Avg. Ride Duration	~12 minutes (Stable & short)	~39 minutes (Long & variable)
Seasonal Impact	Moderate (Sustained baseline in winter)	High (Usage drops dramatically in winter)
Top Station Locations	Commercial & Transit Hubs	Tourist spots & Waterfront Parks

Executed Analysis Queries

Query 1: Statistical Summary of Ride Duration

```sql
SELECT
  member_casual,
  ROUND(AVG(ride_length_minutes), 2) AS avg_duration_min,
  APPROX_QUANTILES(ride_length_minutes, 100)[OFFSET(50)] AS
median_duration_min,
  MAX(ride_length_minutes) AS max_duration_min,
  MIN(ride_length_minutes) AS min_duration_min
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
GROUP BY member_casual;
```

Query 2: Day-of-Week Behavior (Workday vs. Weekend)

```sql
SELECT
  member_casual,
  day_of_week,
  day_name,
  COUNT(*) AS total_trips,
  ROUND(AVG(ride_length_minutes), 2) AS avg_duration_min
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
GROUP BY member_casual, day_of_week, day_name
ORDER BY member_casual, day_of_week;
```

Query 3: Monthly Seasonality Analysis

```sql
SELECT
  member_casual,
  year,
  month,
  month_name,
  COUNT(*) AS total_trips,
  ROUND(AVG(ride_length_minutes), 2) AS avg_duration_min
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
```

```
GROUP BY member_casual, year, month, month_name
ORDER BY year, month, member_casual;
```

Query 4: Top 10 Departure Stations for Casual Riders

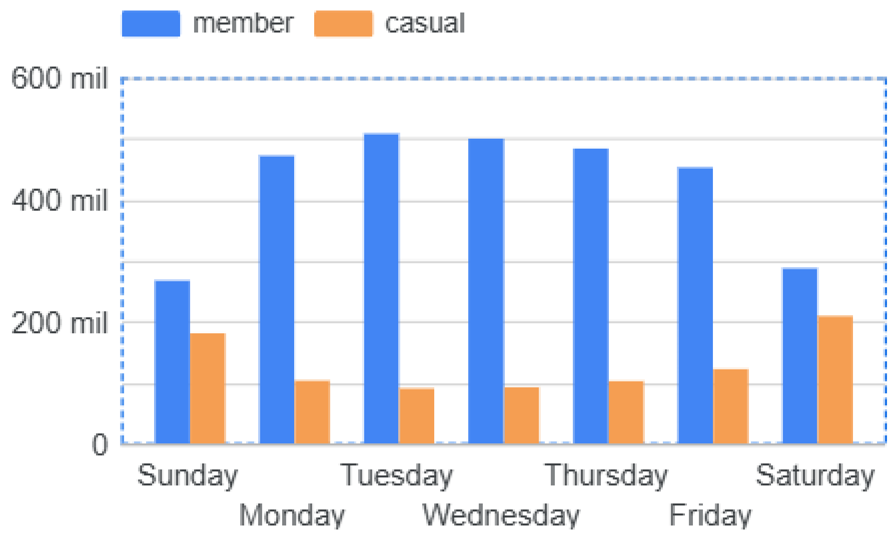
```sql
SELECT
  member_casual,
  start_station_name,
  COUNT(*) AS total_departures
FROM
`project-941e89dd-df86-4a8c-ba4.Cyclistic.cyclistic_12_months_clean`
WHERE member_casual = 'casual'
GROUP BY member_casual, start_station_name
ORDER BY total_departures DESC
LIMIT 10;
```

6. Phase 5: Share

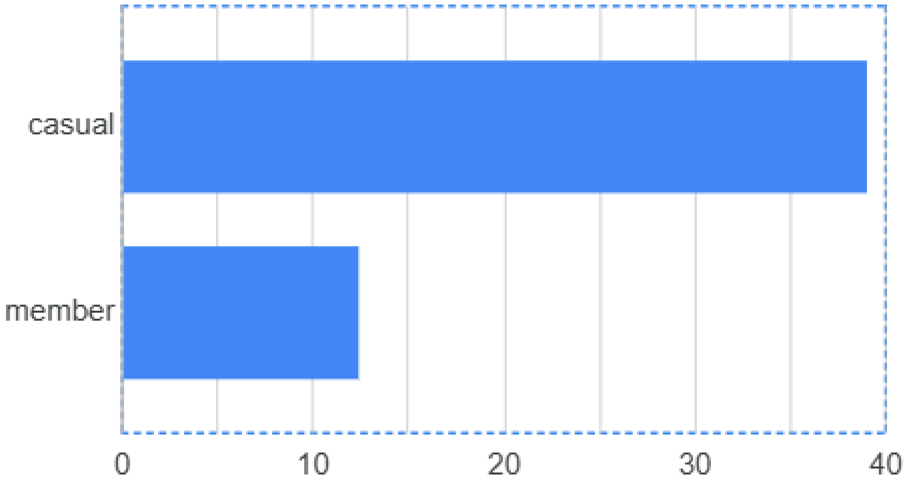
To view the full interactive dashboard, access the [Looker Studio report](#).

Visualization Insights

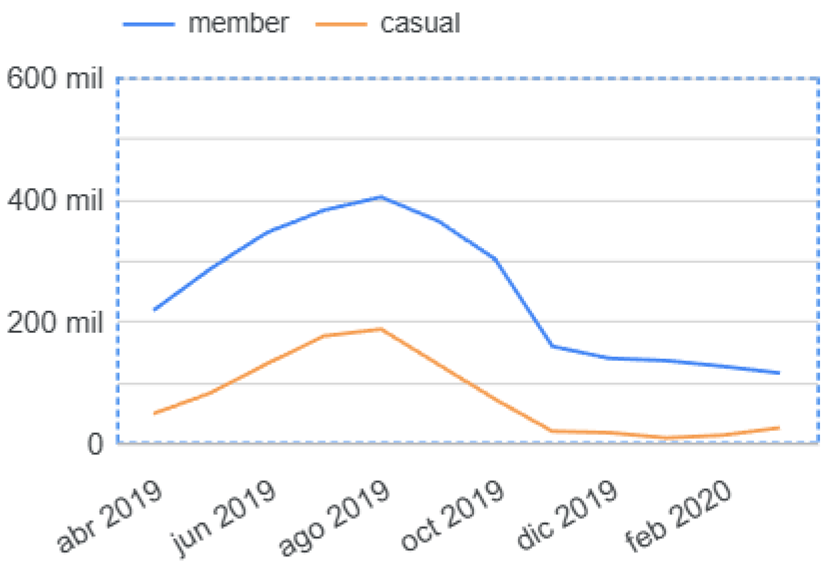
1. **Weekly Ride Volume by User Type:**
    - **Members** peak from Monday through Friday, driven by commuting hours (7:00–9:00 AM and 4:00–6:00 PM).
    - **Casual riders** surge on Saturday and Sunday, showing an inverted usage trend compared to members.



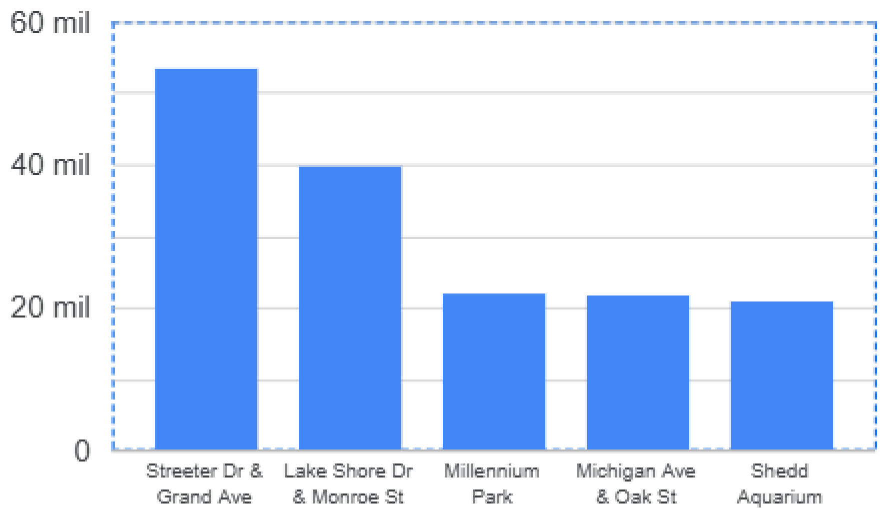
2. **Average Ride Duration:**
    - Casual riders spend **3x longer per ride** (~39 minutes) than members (~12 minutes), proving that casual usage is time-insensitive and leisure-oriented.



3. **Monthly Seasonality:**
    - Casual ride volume expands rapidly between May and September and collapses in winter. Member ride volume declines in winter but maintains a resilient baseline for daily commuting.



4. **Geographic Distribution:**
    - The top departure points for casual riders concentrate at iconic tourist locations: *Streeter Dr & Grand Ave*, *Lake Shore Dr & Monroe St*, *Millennium Park*, *Michigan Ave & Oak St*, and *Shedd Aquarium*.



7. Phase 6: Act

Based on these findings, three marketing and business initiatives are recommended to drive casual-to-member conversions:

Recommendation 1: "Summer Pass to Membership" Credit Program

- **Strategy:** Allow casual riders to apply 100% of single-pass or day-pass purchases made during peak summer months as a direct credit toward an annual membership.

- **Rationale:** Casual riders use the system heavily from May to September. Providing an immediate discount after a summer pass purchase lowers financial friction at the exact moment user intent is highest.

Recommendation 2: Geo-Targeted Campaigns at Tourist Stations

- **Strategy:** Launch physical ads (digital screens, signage) and brand ambassadors targeting the Top 10 casual stations (*Streeter Dr & Grand Ave*, *Lake Shore Dr*, etc.), complemented by mobile geofenced ads.

- **Rationale:** Casual usage is geographically concentrated rather than spread across Chicago. Focusing marketing spend on these specific departure hubs maximizes ROI.

Recommendation 3: Gamified Ride-Time Rewards

- **Strategy:** Introduce a feature in the Cyclistic app that converts ride minutes into "Cyclistic Points," which can be redeemed for annual membership discounts.

- **Rationale:** Since casual riders spend 3x more time per trip (~39 minutes), rewarding duration engages their recreational habits while positioning the annual membership as a cost-effective choice for frequent riders.